METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

Chulalongkorn University · Department of Chemical Engineering

INSTRUMENT LIST

R-201 quench converter and synthesis loop · all P&ID, console and SIS tags

DOCUMENT NO. : MEOH-V1-IN-INL-001

DOCUMENT TITLE : Instrument List

PROJECT : Methanol Converter Ver 1 · OTS

STATUS : Rev 0 · Issued for teaching

Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

Rev.	Date	Revision status	Originator	Checker	Approver
0	27-Oct-26	Issued for teaching	A. Soottitantawat		

1 Purpose and scope

This instrument list (INL) gives one line for every instrument of the Methanol Converter Ver 1 synthesis loop: every tag drawn on the EPC P&IDs MEOH-PID-001 (R-201, E-201, E-204, quench) and MEOH-PID-002 (C-201, E-202, E-203, V-201, make-up, purge and let-down), every tag shown on the simulator console, and the SIS initiators and final elements of the cause and effect diagram MEOH-V1-IN-CE-001 and the SRS MEOH-V1-IN-SRS-001. It lists 66 tags: service, type, calibrated range, normal value at design, alarm and trip set points, I/O type and system, fail position, loop and P&ID sheet.

The trip logic (which cause drives which final element, delays, reset) is not repeated here: see MEOH-V1-IN-CE-001. Sensor voting, response time, SIL and proof test of each safety instrumented function: see MEOH-V1-IN-SRS-001.

2 Basis and assumptions

Item	Basis
Normal value	Simulator model at the design point, noise off (converter feed 1928 kNm³/h, loop 75.0 bar g at V-201, make-up 365.6 kNm³/h, bed inlets 220/225/230/235 °C).
Alarm set points L / H	Simulator alarm list (DCS alarms), identical to the C&E diagram alarm rows.
Trip set points LL / HH	Model trip constants: TSHH-2xx 300 °C (K.tripT), FSLL-201 45 % of design = 868 kNm³/h (K.tripF), LSHH-204 90 % (K.tripL), PSHH-203 85 bar g (K.tripP); delay 5 s.
Calibrated range	Assumed: the model controller span or indicator scale, widened where needed (all converter thermocouples 0–400 °C to cover a cold start-up at 30 °C and the 300 °C trip). Vendor datasheets would confirm.
Flow measurement	Orifice plate + DP transmitter, square root extracted in the DCS (assumed for all flows).
I/O	AI/AO 4–20 mA (HART), DI/DO 24 V DC. Shared transmitter FT-201 wired to the SIS and repeated to the DCS through an isolating splitter; the 12 bed thermocouples wired to the SIS and repeated to the DCS over the SIS–DCS link (assumed).
Fail positions	From the P&ID loop schedule and the model (instrument-air failure): FV-202, PV-203, LV-204, HV-214 fail closed; TV-211, TV-221/231/241 fail open; XV-202 spring-return fail closed.
Spare allowance	20 % installed spare per I/O type, rounded up (typical practice, assumed).

3 Abbreviations

AI/AO analogue input/output · DI/DO digital input/output · BPCS basic process control system (DCS) · SIS safety instrumented system (SIS-201) · FC/FO fail closed/open · L/H low/high alarm · LL/HH low-low/high-high trip · I-n interlock number (C&E) · √ square-root extraction · VSD variable-speed drive · GC gas chromatograph.

4 Instrument list

Tag	Service	Type / device	Range	Unit	Normal	L	H	LL	HH	Trip	I/O	Fail	Loop	P&ID	Note
FIC-201	Converter feed (C-201 discharge circulation)	Orifice FE-201 + DP transmitter FT-201, √ in DCS; PID → C-201 speed	0–2600	kNm³/h	1928	1700	2350				AI BPCS		201	002	Model PID span 0–2600; output = C-201 speed reference SC-201
FSLL-201	Circulation low-low trip (I-2), armed with XV-202 open	SIS function on FT-201 (shared with FIC-201, SRS SIF-2)	0–2600	kNm³/h	1928			868		I-2	AI SIS		201	002	45 % of design (K.tripF); 5 s delay
SC-201 (FIC-201)	C-201 speed reference to the variable-speed drive	DCS output to VSD (loop schedule "C-201 SPEED (SC)")	0–100	%	80						AO BPCS		201	002	Normal = FIC-201 output (speed)

Tag	Service	Type / device	Range	Unit	Normal	L	H	LL	HH	Trip	I/O	Fail	Loop	P&ID	Note
SI-201 (C-201)	C-201 speed feedback	VSD speed signal (not drawn as a bubble)	0–100	%	80						AI BPCS		201	002	
JI-201 (C-201)	C-201 shaft power	VSD power signal (model: calculated)	0–12	MW	6.53						AI BPCS		201	002	Range from the model indicator (0–12 MW); see EQ-ei-05
C-201	Circulator motor run/stop interface	Motor control: run status, start/stop, VSD fault, SIS stop (I-3, I-5)	—		running						DI BPCS DI BPCS DO BPCS DO SIS		201	002	C&E effect S (stop) for LSHH-204 and HS-200
TI-202	C-201 discharge temperature	RTD Pt100 in thermowell + transmitter	0–120	°C	48.6						AI BPCS		202	002	
FIC-202	Make-up syngas from the battery limit	Orifice FE-202 + DP transmitter FT-202, √ in DCS; PID → FV-202	0–600	kNm³/h	365.6	330	400				AI BPCS		202	002	Sets the plant rate
FV-202	Make-up flow control valve	Globe control valve, pneumatic diaphragm, positioner	0–100	% open	50						AO BPCS	FC	202	002	Forced to 0 % by every SIS trip (C&E)
XV-202	Make-up shut-off, final element of I-1 … I-5	On/off valve, spring-return actuator, solenoid XY-202, limit switches	—		open						DO SIS DI SIS DI SIS	FC	202	002	Stroke ≤ 2 s (assumed, SRS); normal: open (SIS RUN)
PI-201	Converter inlet pressure	Pressure transmitter	0–100	bar g	79.7						AI BPCS		201	002	
TI-203	E-202 cold-side outlet = quench gas	RTD Pt100 in thermowell + transmitter	0–200	°C	96.8						AI BPCS		203	002	
TI-252	E-202 hot-side outlet (to E-203)	Thermocouple type K in thermowell + head transmitter	0–250	°C	155.9						AI BPCS		252	002	
TI-253	V-201 separator temperature (E-203 outlet)	RTD Pt100 in thermowell + transmitter	0–120	°C	40		45				AI BPCS		253	002	
TI-254	E-203 cooling water return	RTD Pt100 in thermowell + transmitter	20–80	°C	41.3						AI BPCS		254	002	
PIC-203	Loop pressure at V-201; PID → purge PV-203	Pressure transmitter PT-203	0–100	bar g	75	72	78				AI BPCS		203	002	
PSHH-203	Loop pressure high-high trip (I-4)	SIS pressure transmitter, own tapping (not PT-203), SRS SIF-4	0–100	bar g	75				85	I-4	AI SIS		203	002	5 s delay; PSV-201 set 88 bar g above
PV-203	Purge to reformer fuel header	Globe control valve, pneumatic diaphragm, positioner	0–100	% open	45						AO BPCS	FC	203	002	
FI-203	Purge gas flow to reformer fuel	Orifice FE-203 + DP transmitter FT-203, √ in DCS	0–250	kNm³/h	105.1		150				AI BPCS		203	002	

Tag	Service	Type / device	Range	Unit	Normal	L	H	LL	HH	Trip	I/O	Fail	Loop	P&ID	Note
PSV-201	V-201 / loop relief to flare	Pressure safety valve (relief basis: Relief & vent tab)	—	bar g	closed						—		201	002	Set 88 bar g; no I/O
HV-205	Manual blowdown V-201 → flare	Manual valve (field), restriction orifice TBC	—		closed						—	n/a (manu al)	205	002	Normally closed; field tab hand valve
LIC-204	V-201 crude methanol level; PID → LV-204	Level transmitter LT-204 (DP, remote seals, assumed)	0–100	%	50	25	75				AI BPCS		204	002	
LSHH-204	V-201 level high-high trip (I-3, also stops C-201)	SIS level transmitter, own nozzles (not LT-204), SRS SIF-3	0–100	%	50				90	I-3	AI SIS		204	002	5 s delay
LV-204	Crude methanol let-down to the let-down drum	Angle control valve, anti-cavitation trim (assumed), positioner	0–100	% open	50						AO BPCS	FC	204	002	
FI-209	Crude methanol let-down flow	Orifice FE-209 + DP transmitter FT-209, √ in DCS	0–200	t/h	131.5						AI BPCS		209	002	
XI-210	Methanol production (CH3OH in the crude)	DCS calculated tag (no field device)	0–150	t/h	112.5	105					—		210	—	Soft tag
AI-205 (AIT-205)	Converter inlet CO	Process gas chromatograph AIT-205 (shared, 3 streams of results)	0–10	mol %	4.88						AI BPCS		205	001	
AI-206 (AIT-205)	Converter inlet CO2	AIT-205 (GC)	0–10	mol %	3.35						AI BPCS		205	001	
AI-207 (AIT-205)	Converter inlet CH4 (loop inerts)	AIT-205 (GC)	0–25	mol %	9.93		13				AI BPCS		205	001	
AI-208 (AIT-208)	Converter outlet CH3OH	Process analyser AIT-208 (heated sample line, EQ-ee-04)	0–8	mol %	4.78	3.80					AI BPCS		208	001	
FI-212	Bed 1 feed (main feed through E-201 / TV-211)	Orifice FE-212 + DP transmitter FT-212, √ in DCS	0–1500	kNm³/h	913.5						AI BPCS		212	001	
FI-222	Quench (cold shot) to bed 2	Orifice FE-222 + DP transmitter, √ in DCS	0–800	kNm³/h	244.5						AI BPCS		222	001	
TIC-221	Bed 2 inlet (below the quench distributor); PID → TV-221	Thermocouple type K in thermowell + head transmitter	0–400	°C	225	218	232				AI BPCS		221	001	DCS PID span 100–300 °C (model)
TV-221	Quench valve to bed 2	Butterfly/globe control valve, pneumatic, positioner	0–100	% open	45						AO BPCS	FO	221	001	Fail open: converter goes colder (P&ID note 3)
FI-232	Quench (cold shot) to bed 3	Orifice FE-232 + DP transmitter, √ in DCS	0–800	kNm³/h	393.2						AI BPCS		232	001	
TIC-231	Bed 3 inlet (below the quench distributor); PID → TV-231	Thermocouple type K in thermowell + head transmitter	0–400	°C	230	223	237				AI BPCS		231	001	DCS PID span 100–300 °C (model)

Tag	Service	Type / device	Range	Unit	Normal	L	H	LL	HH	Trip	I/O	Fail	Loop	P&ID	Note
TV-231	Quench valve to bed 3	Butterfly/globe control valve, pneumatic, positioner	0–100	% open	45						AO BPCS	FO	231	001	Fail open: converter goes colder (P&ID note 3)
FI-242	Quench (cold shot) to bed 4	Orifice FE-242 + DP transmitter, √ in DCS	0–800	kNm³/h	376.8						AI BPCS		242	001	
TIC-241	Bed 4 inlet (below the quench distributor); PID → TV-241	Thermocouple type K in thermowell + head transmitter	0–400	°C	235	228	242				AI BPCS		241	001	DCS PID span 100–300 °C (model)
TV-241	Quench valve to bed 4	Butterfly/globe control valve, pneumatic, positioner	0–100	% open	45						AO BPCS	FO	241	001	Fail open: converter goes colder (P&ID note 3)
TIC-211	Bed 1 inlet; PID → TV-211 (E-201 cold-side bypass)	Thermocouple type K in thermowell + head transmitter	0–400	°C	220	212	228				AI BPCS		211	001	DCS PID span 100–300 °C (model)
TV-211	E-201 cold-side bypass	Globe control valve, pneumatic, positioner	0–100	% open	30.7						AO BPCS	FO	211	001	
HIC-214	E-204 start-up heater steam (hand controller)	DCS hand station → HV-214	0–100	%	0						—		214	001	Start-up only; forced to 0 % by every trip
HV-214	E-204 HP steam valve	Globe control valve, pneumatic, positioner	0–100	% open	0						AO BPCS	FC	214	001	C&E effect column names it HIC-214 (EQ-ei-02)
FI-214	E-204 start-up steam flow	Flow element + transmitter on HP steam (to be drawn, EQ-ei-01)	0–60	t/h	0						AI BPCS		214	— (not drawn)	In the model and DCS, not on MEOH-PID-001
TI-205	E-201 cold-side outlet (before the bypass joins)	Thermocouple type K in thermowell + head transmitter	0–320	°C	253.7						AI BPCS		205	001	
TI-251	E-201 hot-side outlet (effluent to E-202)	Thermocouple type K in thermowell + head transmitter	0–320	°C	206.4						AI BPCS		251	001	
TI-212	Bed 1 upper third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	236				300	I-1	AI SIS		2xx	001	
TI-213	Bed 1 lower third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	247.5				300	I-1	AI SIS		2xx	001	
TI-214	Bed 1 exit	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	260		275		300	I-1	AI SIS		2xx	001	TIAH on the bed exit
TI-222	Bed 2 upper third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	247.1				300	I-1	AI SIS		2xx	001	

Tag	Service	Type / device	Range	Unit	Normal	L	H	LL	HH	Trip	I/O	Fail	Loop	P&ID	Note
TI-223	Bed 2 lower third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	265.7				300	I-1	AI SIS		2xx	001	
TI-224	Bed 2 exit	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	276.9		285		300	I-1	AI SIS		2xx	001	TIAH on the bed exit
TI-232	Bed 3 upper third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	248.8				300	I-1	AI SIS		2xx	001	
TI-233	Bed 3 lower third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	262.1				300	I-1	AI SIS		2xx	001	
TI-234	Bed 3 exit	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	270		285		300	I-1	AI SIS		2xx	001	TIAH on the bed exit
TI-242	Bed 4 upper third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	253.1				300	I-1	AI SIS		2xx	001	
TI-243	Bed 4 lower third	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	262.7				300	I-1	AI SIS		2xx	001	
TI-244	Bed 4 exit = converter outlet	Multipoint thermocouple type K (TE-212 … 244), TSHH-2xx input	0–400	°C	267.6		285		300	I-1	AI SIS		2xx	001	TIAH on the bed exit
TSHH-2xx	Any bed thermocouple high-high trip (I-1)	SIS function on the 12 bed thermocouples, 1oo12 (SRS SIF-1)	0–400	°C					300	I-1	—		2xx	001	5 s delay; no I/O of its own
PDI-215	Bed 1 pressure drop	DP transmitter, taps above and below the bed	0–2	bar	0.31						AI BPCS		215	001	
PDI-225	Bed 2 pressure drop	DP transmitter, taps above and below the bed	0–2	bar	0.48						AI BPCS		225	001	
PDI-235	Bed 3 pressure drop	DP transmitter, taps above and below the bed	0–4	bar	0.66		1.30				AI BPCS		235	001	
PDI-245	Bed 4 pressure drop	DP transmitter, taps above and below the bed	0–2	bar	0.88						AI BPCS		245	001	
XI-260	Heat of reaction released in R-201	DCS calculated tag (no field device)	0–120	MW	84.6						—		260	—	Soft tag
HS-200	ESD push-button (I-5)	Push-button on the SIS console, NC contact	—		not pushed					I-5	DI SIS		200	001/002 (SIS table)	No delay; also stops C-201 and E-204 steam

Range: assumed calibrated range. Normal: model at design. Shaded columns LL/HH are SIS trip set points. FI-214 is in the model and the DCS but not drawn on MEOH-PID-001 (EQ-ei-01).

5 I/O count summary

System	Type	Used	Spare 20 % (assumed)	Installed	Channels per card (assumed)	Cards
BPCS	AI	33	7	40	16	3
BPCS	AO	9	2	11	8	2
BPCS	DI	2	1	3	32	1
BPCS	DO	1	1	2	16	1
SIS	AI	15	3	18	16	2
SIS	AO	0	0	0	8	0
SIS	DI	3	1	4	32	1
SIS	DO	2	1	3	16	1
Total		65	16	81		11

Counted from the I/O column of section 4. Soft tags (XI-210, XI-260), PSV-201 and HV-205 have no I/O; SIS functions TSHH-2xx, FSLL-201, PSHH-203, LSHH-204 are logic in SIS-201 on the listed inputs.

6 Control and shut-off valves

Valve	Service	Type (assumed)	Fail	P&ID schedule	Normal opening	Driven by
FV-202	Make-up flow control valve	Globe control valve, pneumatic diaphragm, positioner	FC	FC	50 %	FIC-202
XV-202	Make-up shut-off, final element of I-1 … I-5	On/off valve, spring-return actuator, solenoid XY-202, limit switches	FC	FC (note 2)	open	SIS-201 (I-1 … I-5)
PV-203	Purge to reformer fuel header	Globe control valve, pneumatic diaphragm, positioner	FC	FC	45 %	PIC-203
HV-205	Manual blowdown V-201 → flare	Manual valve (field), restriction orifice TBC	n/a (manual)	manual	closed	operator (field)
LV-204	Crude methanol let-down to the let-down drum	Angle control valve, anti-cavitation trim (assumed), positioner	FC	FC	50 %	LIC-204
TV-221	Quench valve to bed 2	Butterfly/globe control valve, pneumatic, positioner	FO	FO	45 %	TIC-221
TV-231	Quench valve to bed 3	Butterfly/globe control valve, pneumatic, positioner	FO	FO	45 %	TIC-231
TV-241	Quench valve to bed 4	Butterfly/globe control valve, pneumatic, positioner	FO	FO	45 %	TIC-241
TV-211	E-201 cold-side bypass	Globe control valve, pneumatic, positioner	FO	FO	30.7 %	TIC-211
HV-214	E-204 HP steam valve	Globe control valve, pneumatic, positioner	FC	FC	0 %	HIC-214

6.1 Cv check (ISA-75.01 / IEC 60534-2-1 gas sizing, typical practice)

Cv = W / (N6 · Y · √(x · p1 · ρ1)), N6 = 27.3 (W kg/h, p bar a, ρ1 kg/m³); Y = 1 − x / (3 · Fγ · xT); Fγ = γ / 1.40; x limited to Fγ · xT (choked flow); Fp = 1. Rated Cv: the value the model's linear valve law implies at its design opening, rounded up (assumed).

Valve	Flow kNm³/h	W kg/h	P1 / P2 bar a	M kg/kmol	ρ1 kg/m³	x	Choked	Y	Cv req.	Model opening	Rated Cv	Max flow → opening
FV-202	365.6	158,871	87 / 76.01	9.74	31.6	0.126	no	0.940	332.3	50 %	670	415 → 56.3 %
PV-203	105.1	27,337	76.01 / 5	5.83	16.7	0.934	yes	0.667	50.4	45 %	120	150 → 60 %

FV-202: P1 = make-up supply 87.0 bar a (K.Pmu); P2 = C-201 suction = V-201 pressure (model lumps line, orifice and XV-202 into FV-202); max flow 415 kNm³/h (production push). γ = 1.4, Z = 1.03, xT = 0.7 (assumed).

PV-203: P1 = V-201 75.0 bar g; P2 = fuel header 4 bar g (assumed): choked, as the model law Pu = Kpu·v·P1 assumes; max flow = FI-203 high alarm 150 kNm³/h. γ = 1.4, Z = 1.02, xT = 0.7 (assumed).

7 Related documents

Document	Number	Relation to this list
P&ID (EPC)	MEOH-PID-001 / MEOH-PID-002 rev A	Source of the tags, loop schedule and fail positions
Cause and effect diagram	MEOH-V1-IN-CE-001	Trip and alarm logic acting on the tags of this list
Safety requirements specification	MEOH-V1-IN-SRS-001	SIF sensors, voting, response time, SIL, proof test
Electrical load list	MEOH-V1-EE-LL-001	Instrument, DCS and SIS UPS load from the I/O count of section 5
Hazardous-area classification	MEOH-V1-EE-HAC-001	Ex marking of the field instruments (IIC T3, assumed)

8 Engineering queries

No.	To	Issue	Proposed
EQ-ei-01	MEOH-PID-001	FI-214 (E-204 start-up steam flow, 0–60 t/h) is an indicator in the simulator/DCS but no flow element or bubble is drawn on the HP steam line to E-204 on MEOH-PID-001.	Add FE/FT-214 and FI-214 on the HP steam line upstream of HV-214 in MEOH-PID-001 rev B; keep it in the INL.
EQ-ei-02	MEOH-V1-IN-CE-001	The E-204 steam valve is tagged HV-214 on MEOH-PID-001 (FC, driven by HIC-214), but the C&E effect column names the hand controller HIC-214 as the final element.	Name the effect "HV-214 (via HIC-214)" in the C&E rev R1 so the logic, the P&ID and the INL use one valve tag.
EQ-ei-03	MEOH-PID-002 / MEOH-V1-IN-SRS-001	PSHH-203 and LSHH-204 have their own transmitters (SRS SIF-3, SIF-4) but no transmitter tag on MEOH-PID-002; the INL cannot list them as separate devices.	Assign SIS transmitter tags (e.g. a "B" suffix per site convention) on the P&ID and in the INL, with their own process connections.
EQ-ei-04	Simulator model / MEOH-V1-IN-SRS-001	XV-202 is spring-return fail closed (SRS, P&ID note 2), but in the model an instrument-air failure moves only the control valves; XV-202 stays open. FV-202 (FC) still stops the make-up, so the safe state is reached by the BPCS valve, not the SIS valve.	Model XV-202 closing on instrument-air loss (or confirm the actuator design); keep FV-202 FC as the second barrier.
EQ-ei-05	Simulator model (JI-201) / MEOH-V1-EE-LL-001	JI-201 range is 0–12 MW, but the model at 100 % C-201 speed absorbs about 12.6 MW: the indication would saturate at maximum circulation.	Range JI-201 0–16 MW (covers the 14 MW motor rating in MEOH-V1-EE-LL-001).
EQ-ei-06	MEOH-V1-IN-CE-001 (PERM-3)	Start permissive PERM-3 needs every bed thermocouple below 280 °C, but the bed 2–4 exit high alarms are at 285 °C: a restart can be held off by a temperature that raises no alarm.	Review the TI-224/234/244 high alarm (e.g. 278 °C) or add a "PERM-3 not met" message on the SIS screen.
EQ-ei-07	Simulator model (display ranges)	The model shows the bed thermocouples on a 150–320 °C scale and runs the TIC-2x1 PIDs on 100–300 °C, but a cold start-up begins at 30 °C and the trip is 300 °C.	Calibrate all converter thermocouples 0–400 °C (this INL); keep the DCS PID span 100–300 °C as a configuration item.
EQ-ei-08	MEOH-V1-IN-SRS-001	FSLL-201 (SIS) and FIC-201 (BPCS) share FT-201, and TSHH-2xx shares the 12 indicating thermocouples: an FT-201 failure is both an initiating cause and a disabled protection layer.	Follow the SRS proposals (FSLL-201BX on its own FT-201B, TSHH-2oo3 on dedicated elements); add the new transmitters to the INL and the I/O count.